BIOST 2155: Assignment 1

YOUR NAME HERE (YOUR PITT EMAIL HERE)

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Instructions

- This file serves as a template for your assignment. Remember to set `eval = TRUE` below when compiling the report.

```
knitr::opts_chunk$set(echo = TRUE, eval = FALSE,
                      fig.height = 4, fig.width = 7, fig.align = "center",
                      message=FALSE)
```

- Use comments to annotate and explain your code whenever possible.
- This assignment uses one data set: `Boston Housing` and is saved as `assign1_regData.csv`

Theory (10 points)

T1. Mathematical basis of variance-bias tradeoff (10 points)

Assume $Y = f(X) + \epsilon$, where $E(\epsilon) = 0, \text{Var}(\epsilon) = \sigma^2$ and ϵ and $f(X)$ are uncorrelated. Prove the key formula for bias-variance tradeoff:

$$\text{MSE} = (\text{bias})^2 + \text{variance} + \text{irreducible error}$$

where $\text{MSE} = E\left[(y - \hat{f}(x))^2\right]$, $\text{bias} = E[\hat{f}(x)] - f(x)$, $\text{variance} = \text{Var}[\hat{f}(x)]$ and $\text{irreducible error} = \sigma^2$.

Application (Part A): Basics (65 points)

Dataset: `Boston Housing`, saved as `assign1_regData.csv`.

Harrison and Rubinfeld (1978)[1] studied families' willingness to pay for clean air using Boston area housing market data in 1978. The cleaned data consists of 394 rows with **complete** data on census tracts in Boston, with variables collected from a mixture of surveys, administrative records, and other research papers. The variables are described below.

- `CRIM`: per-capita crime rate by town (units and crimes included not specified)
- `ZN`: percentage of residential land zoned for lots over 25,000 sq.ft.
- `INDUS`: proportion of non-retail business acres per town (nonretail businesses include industrial and commercial businesses)
- `CHAS`: Charles River dummy variable (= 1 if tract bounds river; 0 otherwise)
- `NOX`: nitric oxides concentration (parts per 10 million)
- `RM`: average number of rooms per dwelling
- `AGE`: percentage of owner-occupied units built prior to 1940
- `DIS`: weighted log distances to five Boston employment centers
- `RAD`: index of accessibility to radial highways, where a higher number indicates higher accessibility
- `TAX`: full-value property-tax rate per $10,000

[1] David Harrison, Jr and Daniel L Rubinfeld. Hedonic housing prices and the demand for clean air. Journal of Environmental Economics and Management, 5(1):81–102, 1978.

- `PTRATIO`: pupil-teacher ratio by school district
- `B`: $1000(B_k - 0.63)^2$ where B_k is the proportion of Black people by town
- `LSTAT`: % lower status of the population, defined as half of the sum of proportion of adults without some high school education and proportion of male workers classified as laborers
- `MEDV`: Median value of owner-occupied homes, in $1000's

We will load the dataset and set the outcome variable to be `MED`.

```
# Load required libraries and dataset
pacman::p_load(tidyverse, caret, boot)
data <- read_csv("assign1_regData.csv") |> drop_na()
data$CHAS <- as.factor(data$CHAS)
```

A1. Data Exploration & Baseline Model (5 + 5 + 5 = 15 points)

1. Use the `summary` function to provide descriptive statistics for the dataset. Randomly split your data into a training set (80%) and a test set (20%). Make sure that the distribution of the outcome is similar in both sets (please read the documentation for `caret::createDataPartition` to know how). Use a fixed seed (use `seed = 42`) for reproducibility.

```
# your code here
```

2. Perform a brief exploratory data analysis: comment on how the outcome varies with the `RM` predictor - is the behavior expected?

```
# your code here
```

3. Fit a multiple linear regression model using all predictors on the training data. Report the test set Root Mean Squared Error (RMSE).

```
# your code here
```

A2. Resampling and Model Assessment (5 + 8 + 2 = 15 points)

1. Using the training data, perform 10-fold cross-validation on the full multiple linear regression model from A1. What is the average cross-validated RMSE?

```
# your code here
```

2. Use the bootstrap method (with $B = 1000$ resamples, set `seed = 42`) on the training data to estimate the 95% confidence interval (CI) for the `RM` coefficient. Compare the bootstrap-based 95% CI with the 95% CI using the standard formula from our linear regression model.

```
# your code here
```

3. Interpretation: *Write your interpretation of the bootstrap confidence interval here. How does the theoretically-derived interval compare to the one you just estimated with the bootstrap method? If they differ, what could be some potential causes?*

A3. Feature Selection & Regularization (8 + 4 + 4 + 4 = 20 points)

1. Stepwise Selection: On the training data, perform forward stepwise selection to find the best model according to BIC. List the variables selected by the procedure

```
# your code here
```

2. Ridge Regression: Fit a ridge regression model, using 10-fold CV to determine the optimal lambda for the training set. Report this lambda and the test set RMSE.

```
# your code here
```

3. Lasso Regression: Fit a lasso regression model. Use 10-fold CV to determine the optimal lambda. Report optimal lambda, list variables shrunk to zero, and calculate test set RMSE. Finally, plot coefficient paths for varying lambda.

```
# your code here
```

4. Partial Least Squares (PLS): Fit a PLS model. Use 10-fold CV to select the optimal number of components. Report the test set RMSE.

```
# your code here
```

A4. Non-Linear Modeling (5 + 5 + 5 = 15 points)

1. Polynomial Regression: Consider a polynomial regression with `LSTAT` as the only predictor. Use 10-fold CV to select the optimal polynomial degree (1 to 10). Plot how CV errors vary with polynomial degree and the optimal degree. Report test set RMSE for the optimal model.

```
# your code here
```

2. Splines: Fit a natural spline (remember, natural spline = cubic spline + linearity in tails) with `RM` as the predictor. Report test set RMSE.

```
# your code here
```

3. Generalized Additive Model (GAM): Fit three GAMs, each with with natural splines for (a) `LSTAT` alone, (b) `RM` alone, and (c) both `LSTAT` and `RM`. Compare the three models and for the optimal one, report test set RMSE.

```
# your code here
```

Application (Part B): Advanced (25 points)

The `B` variable in this data set is defined as $1000(B_k - 0.63)^2$ where B_k is the proportion of Black people by town. Due to the squared term, this variable treats towns with very low proportions of Black residents and towns with very high proportions similarly, which is a problematic construction. This exercise will investigate how models trained on demographically distinct subgroups can exhibit significant performance bias.

You will train two separate models on two different subgroups of the data and then evaluate their performance on both their own and each other's test sets.

1. Data Partitioning (2 points): First, partition the entire data into two groups based on the median of the `B` variable.
   - Group A: All rows where the `B` value is less than or equal to the median of `B`.
   - Group B: All rows where the `B` value is greater than the median of `B`.

```
# your code here
```

2. Stratified Splitting (8 points): For each group, perform an 80/20 train-test split (remember to set `seed = 42`). Crucially, this split must be stratified by the outcome variable, `MEDV`, to ensure that the distribution of housing prices is similar in both the training and testing sets for each group. This will result in four datasets: `Train A`, `Test A`, `Train B`, and `Test B`.

```
# your code here
```

3. Model Training (10 points): Train two distinct local regression models using the training data from your subgroups:
   - `Model A`: Fit a LOESS model with `span = 0.15` using `Train A` using `RM` and `LSTAT` as predictors.
   - `Model B`: Fit a LOESS model with `span = 0.15` using `Train B` using `RM` and `LSTAT` as predictors.

```
# your code here
```

4. Performance Evaluation and Interpretation (5 points): Calculate the Root Mean Squared Error (RMSE) for the following four scenarios. This will involve evaluating each model on both its "in-group" test set and the "cross-group" test set.
   - In-Group A: The error of `Model A` when predicting `Test A`.
   - In-Group B: The error of `Model B` when predicting `Test B`.
   - Cross-Group $A \to B$: The error of `Model A` when predicting `Test B`.
   - Cross-Group $B \to A$: The error of `Model B` when predicting `Test A`.

Present your four RMSE values in a table. Compare the in-group versus cross-group error rates. What do these results tell you about how the B variable influences model performance? Explain what this implies about the dangers of training and deploying models on demographically specific or unrepresentative data.

```
# your code here
```

Bonus (unscored, will still provide feedback if you answer)

1. LOO-CV: Prove formula (5.2) from ISLR, Section 5.1.2, page 202.
2. Basis of 0.632 bootstrap: Problem 5.4.2 from ISLR, page 219.
3. Continuity and smoothness of cubic splines: Problem 7.9.1 from ISLR, page 321.